1. Brainstorming & Ideation: Multimodal Attendance & Engagement Agent

Project: NeuroClass — Multimodal Attendance & Engagement Agent

Author: Solo Developer Project

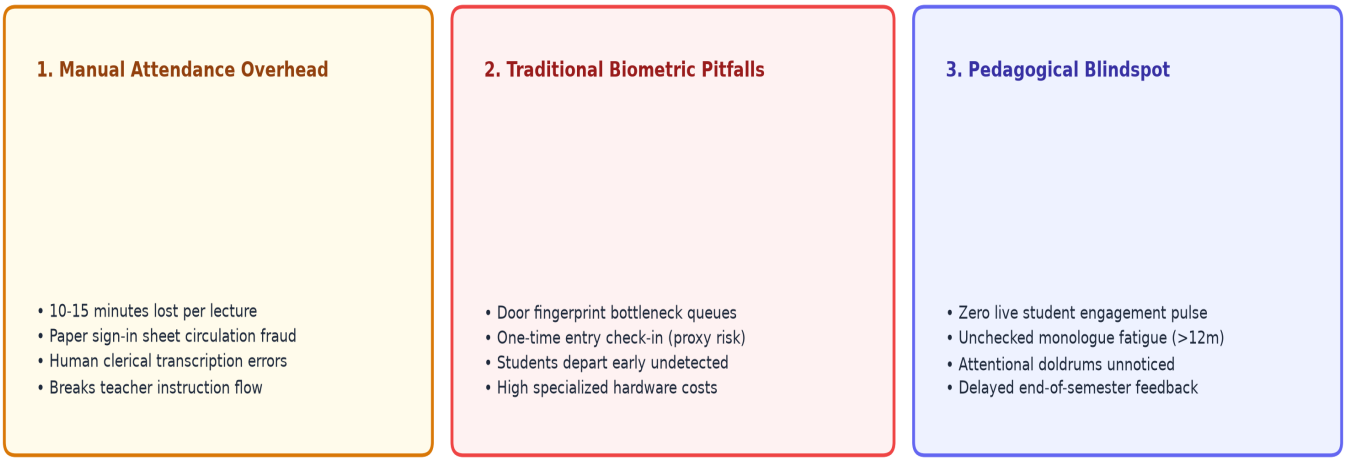
Repository Architecture: Offline-First Edge AI System

Track: AI/ML & Generative AI Track Project

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1. Problem Statement & Background

Traditional educational institutions and modern university lecture halls face a persistent dual challenge: **inefficient administrative attendance tracking** and a **lack of actionable, real-time feedback on student engagement and pedagogical efficacy**.



The Inadequacy of Status Quo Solutions

1. Manual Roll Calls and Paper Sign-in Sheets:

• Consume 15% to 25% of active instructional time.

• High vulnerability to "buddy signing" and proxy attendance.

• Provide zero insight into whether enrolled students remained present or attentive.

2. Fixed Biometric Scanners (Fingerprint / Turnstile RFID):

• Create physical bottlenecks and classroom entry queues.

• Verify presence only at the instant of classroom entry (***t = 0***), failing to account for students who depart immediately after scanning ("ghost attendance").

• Disconnected from classroom pedagogical dynamics.

3. Cloud-Based Video Surveillance & Proctoring:

• Transmit raw camera feeds and classroom audio to remote cloud servers, creating severe data privacy vulnerabilities (violating GDPR and FERPA principles).

• Incur continuous cloud API and GPU rental costs, making widespread institutional adoption cost-prohibitive.

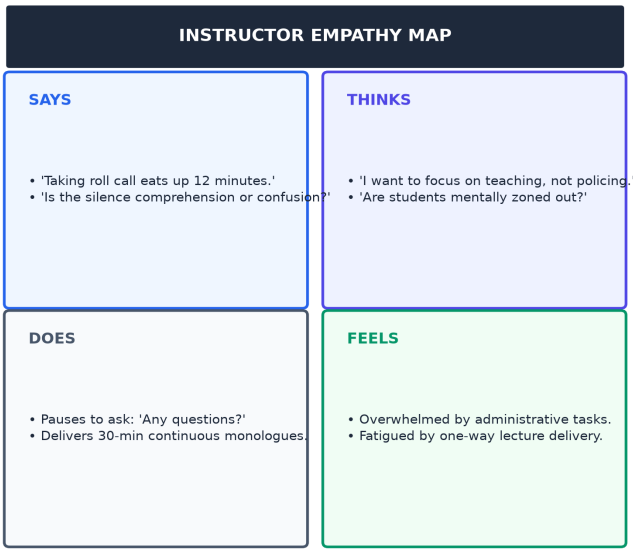
• Invasive speech-to-text eavesdropping compromises student and faculty psychological safety.

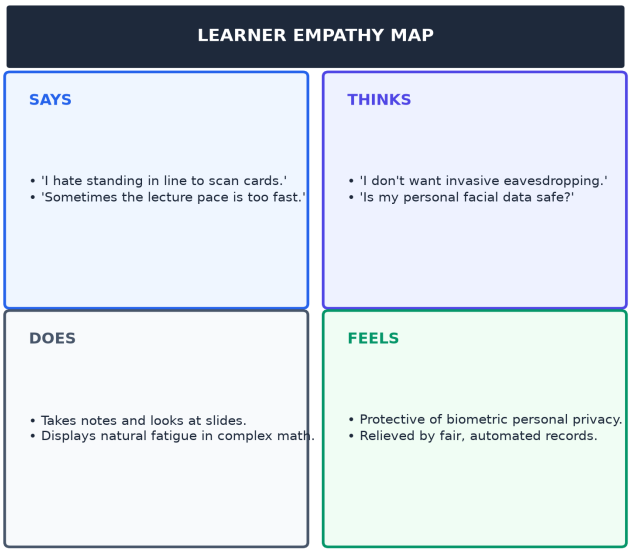
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2. Empathy Maps

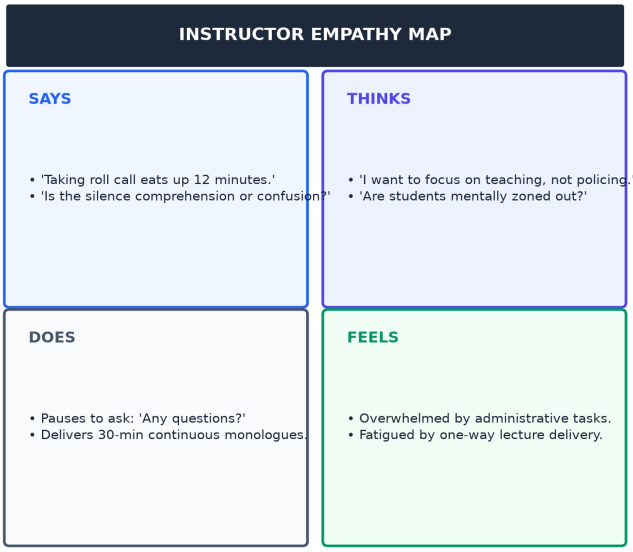
To ground the solution in real human needs, comprehensive empathy mapping was performed for both key classroom stakeholders: the **Course Instructor** and the **University Learner**.

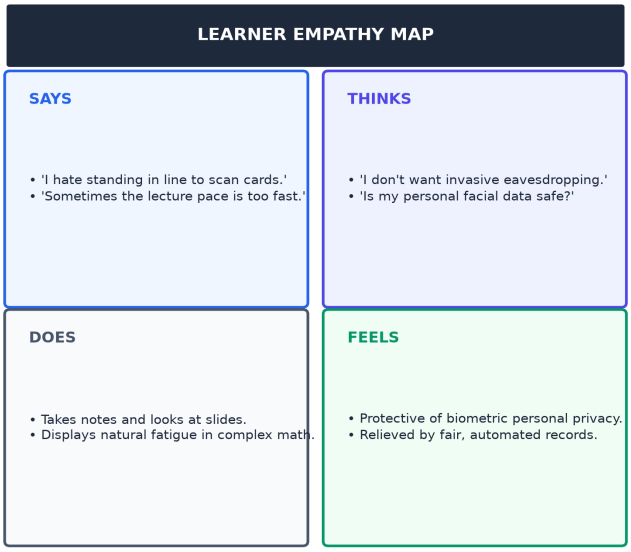
Empathy Map 1: The Course Instructor





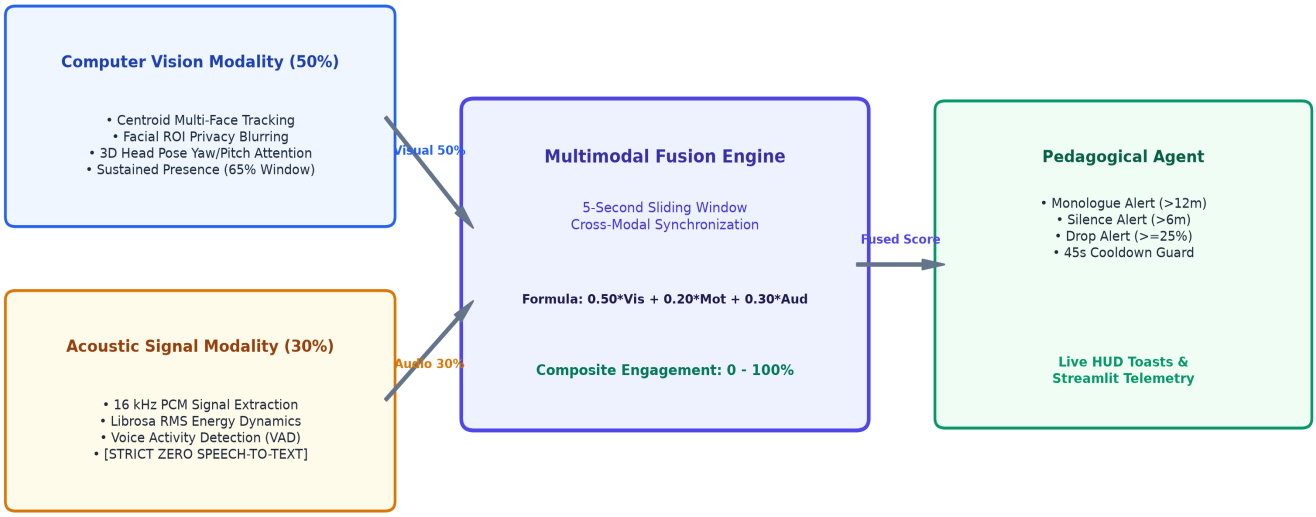
Empathy Map 2: The University Learner





3. Ideation: Multimodal Edge AI Vision & Acoustic Intelligence

To bridge the gap between administrative friction, student privacy, and pedagogical insight, we ideated a **100% Offline Edge Intelligence System** operating on two synchronized sensory modalities:



Key Ideation Innovations

1. Sustained Presence vs. Transient Walk-By Logic:

• Differentiates fleeting visitors (students walking past the open door, duration ***<15*** seconds) from active learners.

• Enforces an adaptive sustained windowing threshold (e.g., student must be actively tracked for ***\ge 65\%*** of the observation period to earn verified attendance).

2. Privacy-Preserving Edge Architecture (Zero Cloud, Zero Speech-to-Text):

• Audio is analyzed strictly through energy, rhythm, pitch variance, and voice activity ratios without running Automatic Speech Recognition (ASR). **No spoken words are ever transcribed or stored.**

• Visual feeds apply optional on-the-fly privacy blurring over facial ROIs.

• Student identities are hashed using salted cryptographic digests (`privacy_salt_offline_classroom_2026`).

3. Multimodal Temporal Fusion & Pedagogical Agent:

• Visual gaze orientation (***50\%***), physical posture/motion balance (***20\%***), and acoustic dialogue dynamics (***30\%***) fuse into a unified composite metric.

• A contextual Rule/LLM Pedagogical Reasoning Agent detects classroom doldrums (e.g., ***>12*** minutes of continuous monologue, ***<35\%*** attention score drop) and generates non-intrusive teaching adjustments.

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4. Core Objectives & Quantitative Success Metrics

Dimension	Target Metric	Engineering Implementation
Instructional Time Saved	***100\%*** attendance automation (***8-12*** mins saved/class)	Automated background face tracking & attendance logging
Walk-By Filtering Accuracy	***<15*** sec detections filtered out	`SustainedPresenceEngine` with sliding window validation
Privacy Certification	***0*** bytes sent to external cloud APIs	Self-contained offline architecture running on standard PC
Acoustic Privacy	***0\%*** word transcription	Energy/VAD/spectral feature extraction via Librosa & NumPy
System Throughput	***\ge 15*** FPS vision, ***<500*** ms audio latency	OpenCV Haar/Centroid tracking + NumPy vectorized math
Pedagogical Actionability	Cooldown-regulated real-time nudges (***45***s)	Context-aware `PedagogicalReasoningAgent`
Identity Flexibility	3-Way Hybrid Enrollment	Image upload, live webcam snap, or automated tracking link

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5. Summary & Transition to Requirements

The ideation phase confirmed that an effective classroom intelligence tool must balance **effortless passive attendance**, **deep non-invasive engagement analytics**, and **absolute student privacy**. These foundational concepts form the direct basis for the engineering specifications detailed in `2.Requirement Analysis`.